

4.5 Referential Integrity (Cascade Update & Delete)

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

CS XI

Chapter 4: Data and Analysis

Lecture 07 of 15

★ Golden

4.5 Referential Integrity (Cascade Update & Delete) ★ Golden

LEARN IT

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it. It stops **orphan records**.

If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student. You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

Parent — STUDENT

ID	Name
101	Ali
102	Sara

Child — RESULT

ID	Grade
101	A
999	F — rejected

Cascade operations

Cascade Update. Change the PK in the parent, and every matching FK in the child changes automatically. Ali's ID 101 → 201 updates Enrollments too. Nothing is left pointing at a vanished 101.

Cascade Delete. Delete the parent, and related child rows are deleted too. If Sara (102) leaves school, her Enrolment and Result rows disappear. That prevents "ghost" marks for a student who is gone.

WORKED EXAMPLE

Before Cascade Update

Student.ID	Enrol.StudentID
101	101

After changing 101 → 201 with Cascade Update ON

Student.ID	Enrol.StudentID
201	201

If Cascade Update were OFF, Access would refuse the change, or (in a sloppy file) leave Enrol stuck on 101 — an orphan.

COMMON MISTAKE

Cascade Delete is powerful and dangerous. Deleting one parent can wipe years of child history. In a real school, you more often **archive** a student (a Status field) than delete them.

EXAM TIP

Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

اردو میں سمجھیں

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔ کیسکیڈ اپ ڈیٹ والدین کی بدلتے پر بچوں کو خود بدل دیتی ہے؛ کیسکیڈ ڈیلیٹ والدین مٹانے پر متعلقہ بچے ریکارڈ بھی مٹا دیتی ہے۔

CHECK YOURSELF

1. What is an orphan record?
2. Cascade Delete: parent Teacher is removed. What happens to that teacher's Class rows?

Answers: 1. A child row whose FK value does not exist in the parent. 2. They are deleted automatically (if Cascade Delete is on).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Referential integrity means:

- a. Every FK matches an existing PK (or is NULL if allowed)
- b. Names are unique
- c. Files are encrypted
- d. Charts are labelled

MCQ 2. Cascade update:

- a. Changes child FKs when the parent PK changes
- b. Deletes the DBMS
- c. Sorts tables
- d. Draws ER

MCQ 3. Cascade delete:

- a. Deletes child rows when the parent is deleted
- b. Inserts a parent
- c. Creates a view
- d. Opens a form

MCQ 4. Inserting an FK that does not exist should:

- a. Be rejected
- b. Be auto-sorted
- c. Draw a pie
- d. Start Agile

MCQ 5. Orphan records are:

- a. Child rows whose parent is gone
- b. Primary keys
- c. Reports
- d. Queries

MCQ 6. Referential integrity is enforced by the:

- a. DBMS
- b. Monitor
- c. Mouse
- d. Compiler of Python only

Extra short questions

1. Define referential integrity.
2. What does cascade update solve?
3. When is cascade delete dangerous?
4. Give an invalid operation.

Extra long question

1. In Members-Borrow, specify restrict vs cascade for UPDATE and DELETE with a Sindh-college story. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. FK values must exist in the parent PK (unless NULL is allowed).
2. If roll number 12 becomes 112, all Marks.roll 12 become 112 automatically.
3. Deleting a member could wipe years of borrow history — sometimes restrict instead.
4. INSERT borrow for MemberID 999 when no such member.

Long

1. UPDATE MemberID: cascade so old cards still match. DELETE member who still has books: restrict (cannot delete). DELETE member with no books: allow. Show one rejected SQL/Access action.